

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

• **Analytical:** Data Analysis, Exploratory Data Analysis, Statistical Analysis, Trend Analysis, Root Cause Analysis, KPI Reporting, Data Validation, Data Cleaning, Anomaly Detection, ETL, Predictive Analytics

• **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j

• **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization

• **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

• Owned end-to-end design and development of **cross-platform system integrating** wearable data for 100+ users, enabling scalable **data ingestion** and analytics

• Designed and optimized **PostgreSQL architecture** for high-volume, production-scale datasets, improving query performance by 38%

• Built robust, production-ready **Python** pipeline with retrieval and context summarization, improving multi-turn response accuracy by 43%

• Developed **backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on reliability and scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Designed and deployed highly **scalable NLP pipelines in Python** to process high-volume datasets, reducing processing latency by 41%

• Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing and pattern detection

• Developed production-grade **APIs** and deployment workflows to improve system reliability and reproducibility

• Enabled large-scale data processing across social media streams using **distributed system architecture** principles

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

• Built and deployed scalable, microservices-based Retrieval-Augmented Generation (**RAG**) system using **FastAPI**, **PostgreSQL**, and vector embeddings

• Designed efficient, fault-tolerant **ETL** pipelines to process and index large document datasets for low-latency retrieval

• Improved system performance by 64% through optimized indexing strategies and pipeline efficiency improvements

• Developed backend services enabling real-time, high-performance querying of unstructured data

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

• Built interactive Power BI and Tableau dashboards for logistics KPIs, reducing reporting time by 40% and improving stakeholder visibility into operational performance across business units

• Collected, cleaned, and transformed large-scale operational datasets using Python and PySpark to identify failure patterns, trends, and performance issues across business operations

• Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data quality issues, speeding root-cause analysis by 26%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

• Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, applying **SQL**, **DAX**, and **statistical analysis** to surface KPI trends and support cross-functional business decision-making

• Developed automated data pipelines and reporting workflows using Excel and VBA, implementing validation and scheduling logic, reducing report preparation time by 42% and improving data consistency

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

• Led cross-functional team of **6** to deliver product updates for DeepCover, game, using user feedback and performance insights to guide data-driven feature decisions

• Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, improving cross-team visibility and supporting on-time milestone reviews

PROJECTS

FMCG Sales Demand Forecasting

March 2025

View Project

• Analyzed FMCG sales data across categories to identify trends, seasonal patterns, and key performance indicators, documenting analytical methodologies and findings for stakeholder reference

• Built and compared forecasting models including **ARIMA**, **SARIMA**, and regression to evaluate demand patterns and improve forecast accuracy using RMSE and MAPE

• Derived actionable insights on promotions, pricing, and stock behavior, enabling better inventory planning and reducing stockouts and overstocking risks